

Experiment No

PTo study and understand the concepts of morphology in Natural Language Processing through the implementation and analysis of Add-Delete tables for word transformation.

Install & Import liabraries

```
import pandas as pd

# -----
# Function for Morphological Rule
# -----
def add_delete_rule(base, add="", delete_suffix=""):
    """
    base = original word
    add = string to add
    delete_suffix = string to remove from end
    """
    if delete_suffix and base.endswith(delete_suffix):
        new_word = base[:-len(delete_suffix)] + add
    else:
        new_word = base + add
    return new_word

# -----
# Examples
# -----
examples = [
    ("play", "ing", ""),
    ("walk", "ed", ""),
    ("cat", "s", ""),
    ("try", "ies", "y"),
    ("happy", "ness", ""),
    ("legal", "illegal", ""), # special case
]

# -----
# Create Add-Delete Table
# -----
data = []

for base, add, delete in examples:
    result = add_delete_rule(base, add, delete)
    data.append([base, add, delete if delete else "-", result])

df = pd.DataFrame(data, columns=["Base Word", "Add", "Delete", "Result Word"])

print(df)
```

	Base Word	Add	Delete	Result Word
0	play	ing	-	playing
1	walk	ed	-	walked
2	cat	s	-	cats
3	try	ies	y	tries
4	happy	ness	-	happyness
5	legal	illegal	-	legalillegal

```
def morphology(base, prefix="", suffix="", delete_suffix=""):
    if delete_suffix and base.endswith(delete_suffix):
        base = base[:-len(delete_suffix)]
    return prefix + base + suffix

examples2 = [
    ("happy", "un", "", ""),
    ("possible", "im", "", ""),
    ("regular", "ir", "", ""),
    ("use", "", "ful", ""),
    ("act", "", "ion", "")
]

data2 = []

for base, prefix, suffix, delete in examples2:
    result = morphology(base, prefix, suffix, delete)
    data2.append([base, prefix if prefix else "-",
                  suffix if suffix else "-",
                  result])
```

```
        delete if delete else "-",
        result])

df2 = pd.DataFrame(data2,
                    columns=["Base", "Prefix Add", "Suffix Add", "Delete", "Result"])

print(df2)
```

	Base	Prefix Add	Suffix Add	Delete	Result
0	happy	un	-	-	unhappy
1	possible	im	-	-	impossible
2	regular	ir	-	-	irregular
3	use	-	ful	-	useful
4	act	-	ion	-	action

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